

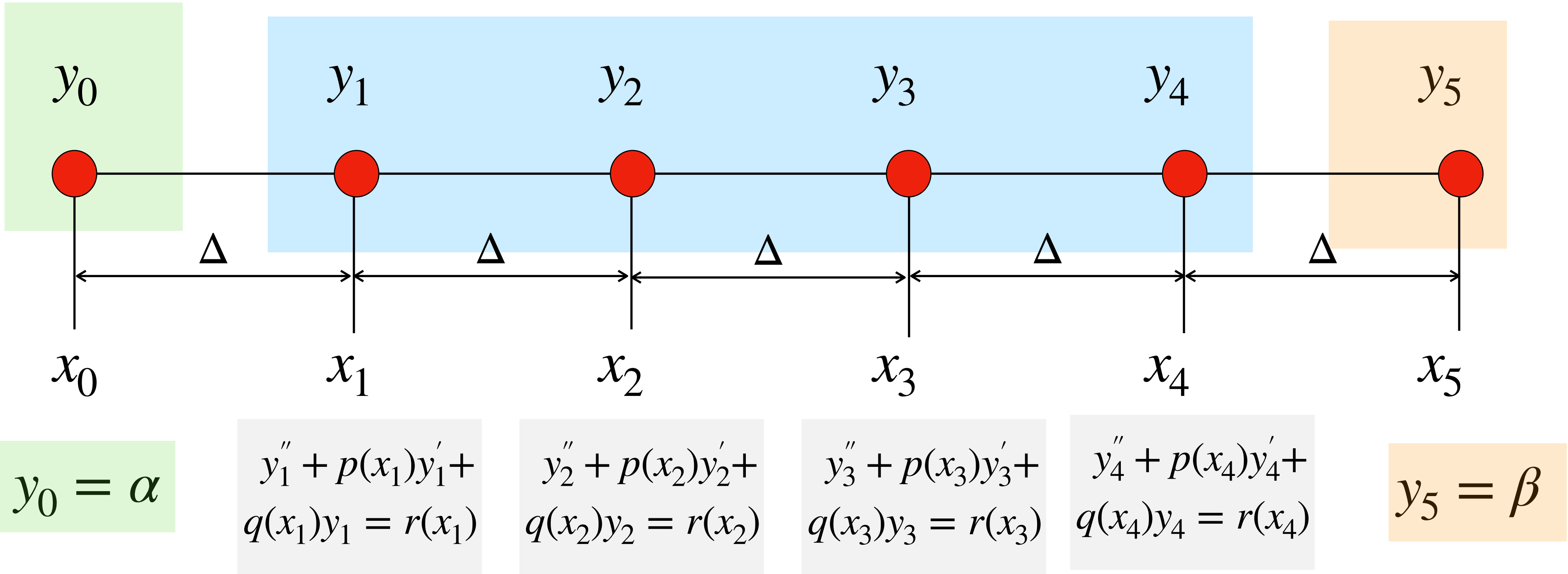
**Boundary Value Problem:
Finite Difference Approach
(Dirichlet Boundary Conditions)**

First consider the case where you “discretise” your domain $a \leq x \leq b$ into 5 intervals.

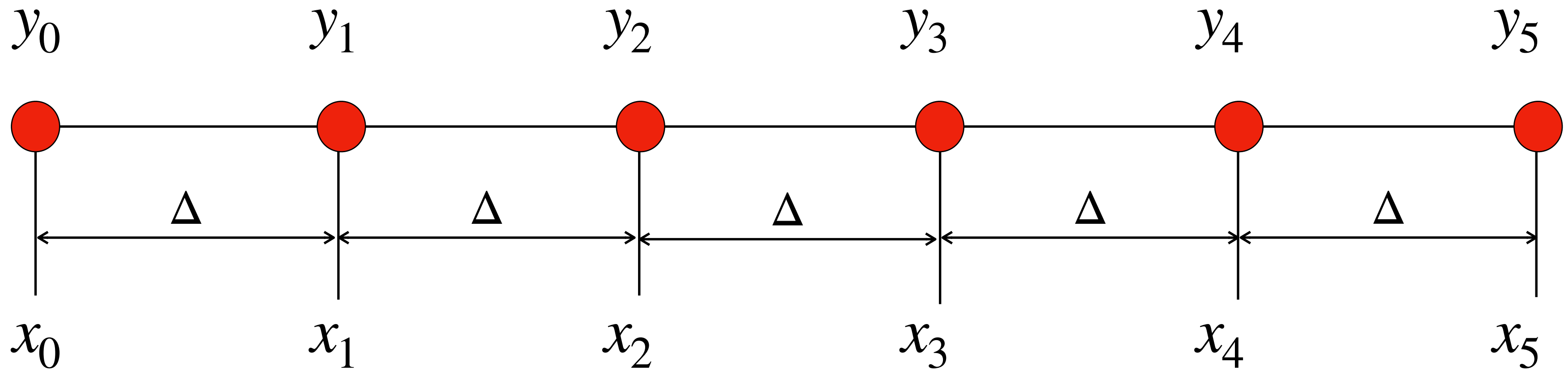
$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = r(x)$$

$$\begin{aligned} \bar{y}(a) = y_0 = \alpha \\ y(b) = y_5 = \beta \end{aligned}$$

Values of y given at the boundaries:
Dirichlet boundary conditions.



So you need to solve for the unknowns y_1, y_2, y_3 and y_4 .



$$y_0 = \alpha \quad \left[\begin{array}{cccc} y_1'' + p(x_1)y_1' + & y_2'' + p(x_2)y_2' + & y_3'' + p(x_3)y_3' + & y_4'' + p(x_4)y_4' + \\ q(x_1)y_1 = r(x_1) & q(x_2)y_2 = r(x_2) & q(x_3)y_3 = r(x_3) & q(x_4)y_4 = r(x_4) \end{array} \right] \quad y_5 = \beta$$

Using
Central difference
approximation

$$\begin{aligned} y_1'' &= \frac{y_2 - 2y_1 + y_0}{\Delta^2} \\ y_2'' &= \frac{y_3 - 2y_2 + y_1}{\Delta^2} \\ y_3'' &= \frac{y_4 - 2y_3 + y_2}{\Delta^2} \\ y_4'' &= \frac{y_5 - 2y_4 + y_3}{\Delta^2} \end{aligned}$$

$$\begin{Bmatrix} y_1'' \\ y_2'' \\ y_3'' \\ y_4'' \end{Bmatrix} + \begin{Bmatrix} p(x_1)y_1' \\ p(x_2)y_2' \\ p(x_3)y_3' \\ p(x_4)y_4' \end{Bmatrix} + \begin{Bmatrix} q(x_1)y_1 \\ q(x_2)y_2 \\ q(x_3)y_3 \\ q(x_4)y_4 \end{Bmatrix} = \begin{Bmatrix} r(x_1) \\ r(x_2) \\ r(x_3) \\ r(x_4) \end{Bmatrix}$$

Using Central difference
approximation

$p(x_1)y_1' = p(x_1)\frac{y_2 - y_0}{2\Delta}$	$q(x_1)y_1$	$r(x_1)$
$p(x_2)y_2' = p(x_2)\frac{y_3 - y_1}{2\Delta}$	$q(x_2)y_2$	$r(x_2)$
$p(x_3)y_3' = p(x_3)\frac{y_4 - y_2}{2\Delta}$	$q(x_3)y_3$	$r(x_3)$
$p(x_4)y_4' = p(x_4)\frac{y_5 - y_3}{2\Delta}$	$q(x_4)y_4$	$r(x_4)$

y_0 y_1 y_2 y_3 y_4 y_5

x_0 x_1 x_2 x_3 x_4 x_5

$y_0 = \alpha$ $y_1'' + p(x_1)y_1' + q(x_1)y_1 = r_1$ $y_2'' + p(x_2)y_2' + q(x_2)y_2 = r_2$ $y_3'' + p(x_3)y_3' + q(x_3)y_3 = r_3$ $y_4'' + p(x_4)y_4' + q(x_4)y_4 = r_4$ $y_5 = \beta$

$$\begin{Bmatrix} y_1'' \\ y_2'' \\ y_3'' \\ y_4'' \end{Bmatrix} + \begin{Bmatrix} p(x_1)y_1' \\ p(x_2)y_2' \\ p(x_3)y_3' \\ p(x_4)y_4' \end{Bmatrix} + \begin{Bmatrix} q(x_1)y_1 \\ q(x_2)y_2 \\ q(x_3)y_3 \\ q(x_4)y_4 \end{Bmatrix} = \begin{Bmatrix} r(x_1) \\ r(x_2) \\ r(x_3) \\ r(x_4) \end{Bmatrix}$$

$$\begin{aligned} y_1'' &= \frac{y_2 - 2y_1 + y_0}{\Delta^2} \\ y_2'' &= \frac{y_3 - 2y_2 + y_1}{\Delta^2} \\ y_3'' &= \frac{y_4 - 2y_3 + y_2}{\Delta^2} \\ y_4'' &= \frac{y_5 - 2y_4 + y_3}{\Delta^2} \end{aligned}$$

$$\begin{aligned} p(x_1)y_1' &= p(x_1)\frac{y_2 - y_0}{2\Delta} \\ p(x_2)y_2' &= p(x_2)\frac{y_3 - y_1}{2\Delta} \\ p(x_3)y_3' &= p(x_3)\frac{y_4 - y_2}{2\Delta} \\ p(x_4)y_4' &= p(x_4)\frac{y_5 - y_3}{2\Delta} \end{aligned}$$

$$\frac{1}{\Delta^2} \begin{bmatrix} 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{bmatrix} \begin{Bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix} + \frac{1}{\Delta} \begin{bmatrix} -\frac{1}{2}p(x_1) & 0 & \frac{1}{2}p(x_1) & 0 & 0 & 0 \\ 0 & -\frac{1}{2}p(x_2) & 0 & \frac{1}{2}p(x_2) & 0 & 0 \\ 0 & 0 & -\frac{1}{2}p(x_3) & 0 & \frac{1}{2}p(x_3) & 0 \\ 0 & 0 & 0 & -\frac{1}{2}p(x_4) & 0 & \frac{1}{2}p(x_4) \end{bmatrix} \begin{Bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix} + \begin{bmatrix} 0 & q(x_1) & 0 & 0 & 0 & 0 \\ 0 & 0 & q(x_2) & 0 & 0 & 0 \\ 0 & 0 & 0 & q(x_3) & 0 & 0 \\ 0 & 0 & 0 & 0 & q(x_4) & 0 \end{bmatrix} \begin{Bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix} = \begin{Bmatrix} r(x_1) \\ r(x_2) \\ r(x_3) \\ r(x_4) \end{Bmatrix}$$

$$\frac{1}{\Delta^2} \begin{bmatrix} 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{bmatrix} \begin{Bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix} + \frac{1}{\Delta} \begin{bmatrix} -\frac{1}{2}p(x_1) & 0 & \frac{1}{2}p(x_1) & 0 & 0 & 0 \\ 0 & -\frac{1}{2}p(x_2) & 0 & \frac{1}{2}p(x_2) & 0 & 0 \\ 0 & 0 & -\frac{1}{2}p(x_3) & 0 & \frac{1}{2}p(x_3) & 0 \\ 0 & 0 & 0 & -\frac{1}{2}p(x_4) & 0 & \frac{1}{2}p(x_4) \end{bmatrix} \begin{Bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix} + \begin{bmatrix} 0 & q(x_1) & 0 & 0 & 0 & 0 \\ 0 & 0 & q(x_2) & 0 & 0 & 0 \\ 0 & 0 & 0 & q(x_3) & 0 & 0 \\ 0 & 0 & 0 & 0 & q(x_4) & 0 \end{bmatrix} \begin{Bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix} = \begin{Bmatrix} r(x_1) \\ r(x_2) \\ r(x_3) \\ r(x_4) \end{Bmatrix}$$

$$y_0 = \alpha$$

$$y_5 = \beta$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{\Delta^2} - \frac{p(x_1)}{2\Delta} & -\frac{2}{\Delta^2} + q(x_1) & \frac{1}{\Delta^2} + \frac{p(x_1)}{2\Delta} & 0 & 0 & 0 \\ 0 & \frac{1}{\Delta^2} - \frac{p(x_2)}{2\Delta} & -\frac{2}{\Delta^2} + q(x_2) & \frac{1}{\Delta^2} + \frac{p(x_2)}{2\Delta} & 0 & 0 \\ 0 & 0 & \frac{1}{\Delta^2} - \frac{p(x_3)}{2\Delta} & -\frac{2}{\Delta^2} + q(x_3) & \frac{1}{\Delta^2} + \frac{p(x_3)}{2\Delta} & 0 \\ 0 & 0 & 0 & \frac{1}{\Delta^2} - \frac{p(x_4)}{2\Delta} & -\frac{2}{\Delta^2} + q(x_4) & \frac{1}{\Delta^2} + \frac{p(x_4)}{2\Delta} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix} = \begin{Bmatrix} \alpha \\ r(x_1) \\ r(x_2) \\ r(x_3) \\ r(x_4) \\ \beta \end{Bmatrix}$$

This is now a 6x6 set of linear algebraic system of equations. You can use any of the technique in lectures 5 and 6 to solve for

$$\begin{Bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix}$$

If we define

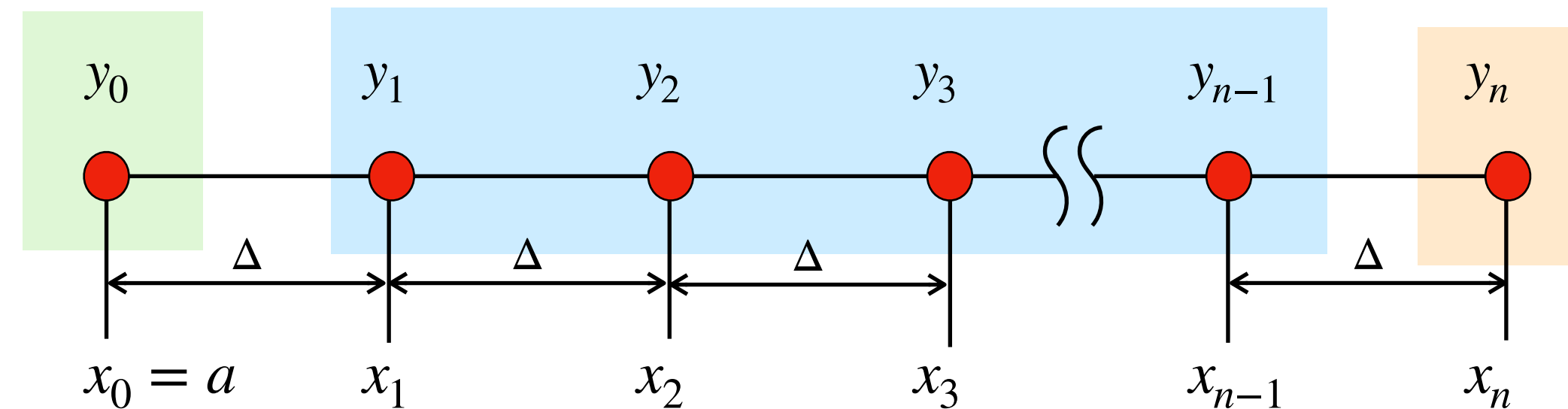
$$f_i = \frac{1}{\Delta^2} - \frac{p(x_i)}{2\Delta} \quad g_i = -\frac{2}{\Delta^2} + q(x_i) \quad h_i = \frac{1}{\Delta^2} + \frac{p(x_i)}{2\Delta}$$

Then Eq. (14.2) can be written as

Tridiagonal matrix. Easy to solve. $\longrightarrow$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ f_1 & g_1 & h_1 & 0 & 0 & 0 \\ 0 & f_2 & g_2 & h_2 & 0 & 0 \\ 0 & 0 & f_3 & g_3 & h_3 & 0 \\ 0 & 0 & 0 & f_4 & g_4 & h_4 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix} = \begin{Bmatrix} \alpha \\ r(x_1) \\ r(x_2) \\ r(x_3) \\ r(x_4) \\ \beta \end{Bmatrix}$$

The above matrix system was for 5 intervals. In general if you divide a domain up into n intervals,



then the system of equations that you need to solve would look like

$$\begin{bmatrix} 1 & 0 & \dots & \dots & \dots & \dots & 0 \\ f_1 & g_1 & h_1 & 0 & \dots & \dots & 0 \\ 0 & f_2 & g_2 & h_2 & 0 & \dots & 0 \\ 0 & 0 & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & \ddots & f_{n-2} & g_{n-2} & h_{n-2} & 0 \\ 0 & 0 & 0 & \ddots & f_{n-1} & g_{n-1} & h_{n-1} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_{n-2} \\ y_{n-1} \\ y_n \end{Bmatrix} = \begin{Bmatrix} \alpha \\ r(x_1) \\ r(x_2) \\ \vdots \\ r(x_{n-2}) \\ r(x_{n-1}) \\ \beta \end{Bmatrix}$$

Solve linear BVP using finite difference. Taken from “Applied Numerical Methods Using MATLAB” by Yang, Cao, Chung and Morris

$$y''(x) + \frac{2}{x}y'(x) - \frac{2}{x^2}y(x) = 0$$

$x \in [1,2]$ and the boundary conditions $y(1) = 5$ and $y(2) = 3$. Use equally spaced data points. Compare your solution with the exact expression for

$$y(x) = x + \frac{4}{x^2}$$

$$y''(x) + \frac{2}{x}y'(x) - \frac{2}{x^2}y(x) = 0$$

$$\frac{d^2y}{dx^2} \approx \frac{y_{i-1} - 2y_i + y_{i+1}}{\Delta^2}$$

$$\frac{2}{x} \frac{dy}{dx} \approx \frac{2}{x_i} \frac{y_{i+1} - y_{i-1}}{2\Delta}$$

$$-\frac{2}{x^2}y \approx -\frac{2}{x_i^2}y_i$$

If you split your domain up into 5 intervals, and put your equations in matrix form, they will look like

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ -1/(\Delta x_2) + 1/\Delta^2 & -2/\Delta^2 - 2/x_2^2 & 1/(\Delta x_2) + 1/\Delta^2 & 0 & 0 & 0 \\ 0 & -1/(\Delta x_3) + 1/\Delta^2 & -2/\Delta^2 - 2/x_3^2 & 1/(\Delta x_3) + 1/\Delta^2 & 0 & 0 \\ 0 & 0 & -1/(\Delta x_4) + 1/\Delta^2 & -2/\Delta^2 - 2/x_4^2 & 1/(\Delta x_4) + 1/\Delta^2 & 0 \\ 0 & 0 & 0 & -1/(\Delta x_5) + 1/\Delta^2 & -2/\Delta^2 - 2/x_5^2 & 1/(\Delta x_5) + 1/\Delta^2 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{Bmatrix} = \begin{Bmatrix} 5 \\ 0 \\ 0 \\ 0 \\ 0 \\ 3 \end{Bmatrix}$$

$$y''(x) + \frac{2}{x}y'(x) - \frac{2}{x^2}y(x) = 0$$

$$\frac{d^2y}{dx^2} \approx \frac{y_{i-1} - 2y_i + y_{i+1}}{\Delta^2} \quad \frac{2}{x} \frac{dy}{dx} \approx \frac{2}{x_i} \frac{y_{i+1} - y_{i-1}}{2\Delta} \quad -\frac{2}{x^2}y \approx -\frac{2}{x_i^2}y_i$$

If you split your domain up into 5 intervals, and put your equations in matrix form, they will look like

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ -1/(\Delta x_2) + 1/\Delta^2 & -2/\Delta^2 - 2/x_2^2 & 1/(\Delta x_2) + 1/\Delta^2 & 0 & 0 & 0 \\ 0 & -1/(\Delta x_3) + 1/\Delta^2 & -2/\Delta^2 - 2/x_3^2 & 1/(\Delta x_3) + 1/\Delta^2 & 0 & 0 \\ 0 & 0 & -1/(\Delta x_4) + 1/\Delta^2 & -2/\Delta^2 - 2/x_4^2 & 1/(\Delta x_4) + 1/\Delta^2 & 0 \\ 0 & 0 & 0 & -1/(\Delta x_5) + 1/\Delta^2 & -2/\Delta^2 - 2/x_5^2 & 1/(\Delta x_5) + 1/\Delta^2 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{Bmatrix} = \begin{Bmatrix} 5 \\ 0 \\ 0 \\ 0 \\ 0 \\ 3 \end{Bmatrix}$$

The boundary conditions will give you $y_0 = 5$ and $y_5 = 3$